

*LOCATION SYNC

Location-aware remote content management for moving and static displays

Blackbox hackathon problem statement 2



Right content.
Right place.

...

THE PROBLEM

Many displays, many places, some of them moving

SCALE

manual updates
do not scale

LOCATION

content must
follow the
display

CONTROL

someone
approves what
goes live;
advertisers want
proof it played



OUR SOLUTION

ADVERTISER PORTAL

upload · target
zones/devices ·
day and time
window · play
report

 **GEOCAST**

ADVERTISER PORTAL

 Dashboard

 **Campaigns**

 Profile

 **Sundaram Yadav**
1231232

 Sign Out

Campaigns

Manage, preview, and track your advertising campaigns

 **Cards**

 Table

 **New Campaign**



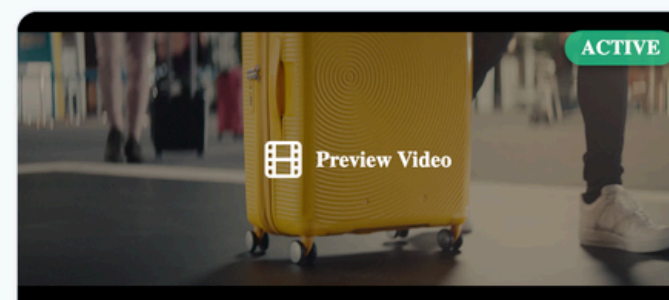
Kurkure

U.P. Ka No.1 Taste | Kurkure

 Zones: 1,9

 20/09/2026 → 23/09/2026

[Manage Campaign →](#)



American Tourister

Cristiano Ronaldo with American Tourister

 Zones: 8,13,12

 20/09/2026 → 21/10/2026

[Manage Campaign →](#)



car photo

car photo

 Zones: 1,8,13

 21/09/2026 → 24/09/2026

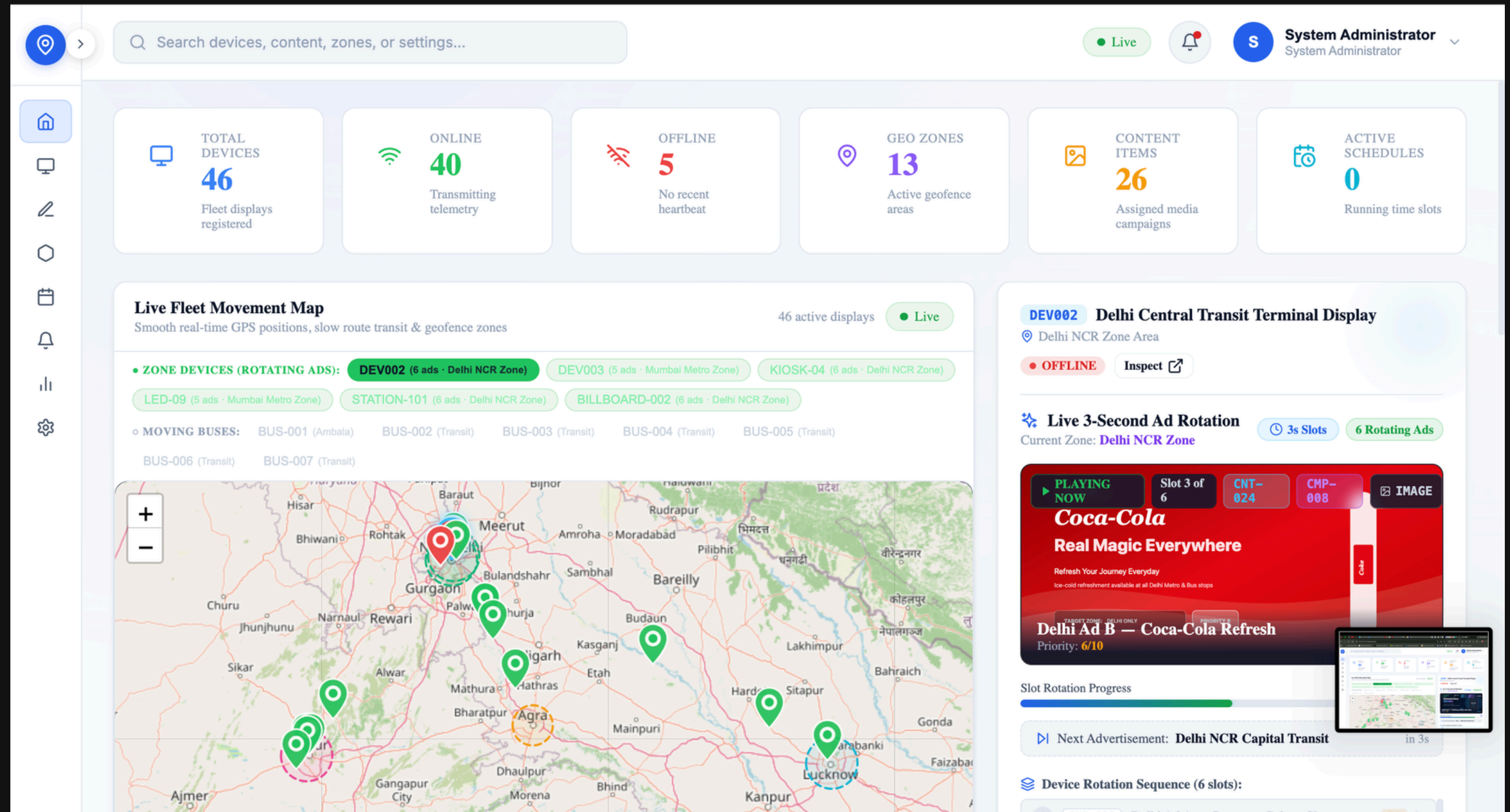
[Manage Campaign →](#)



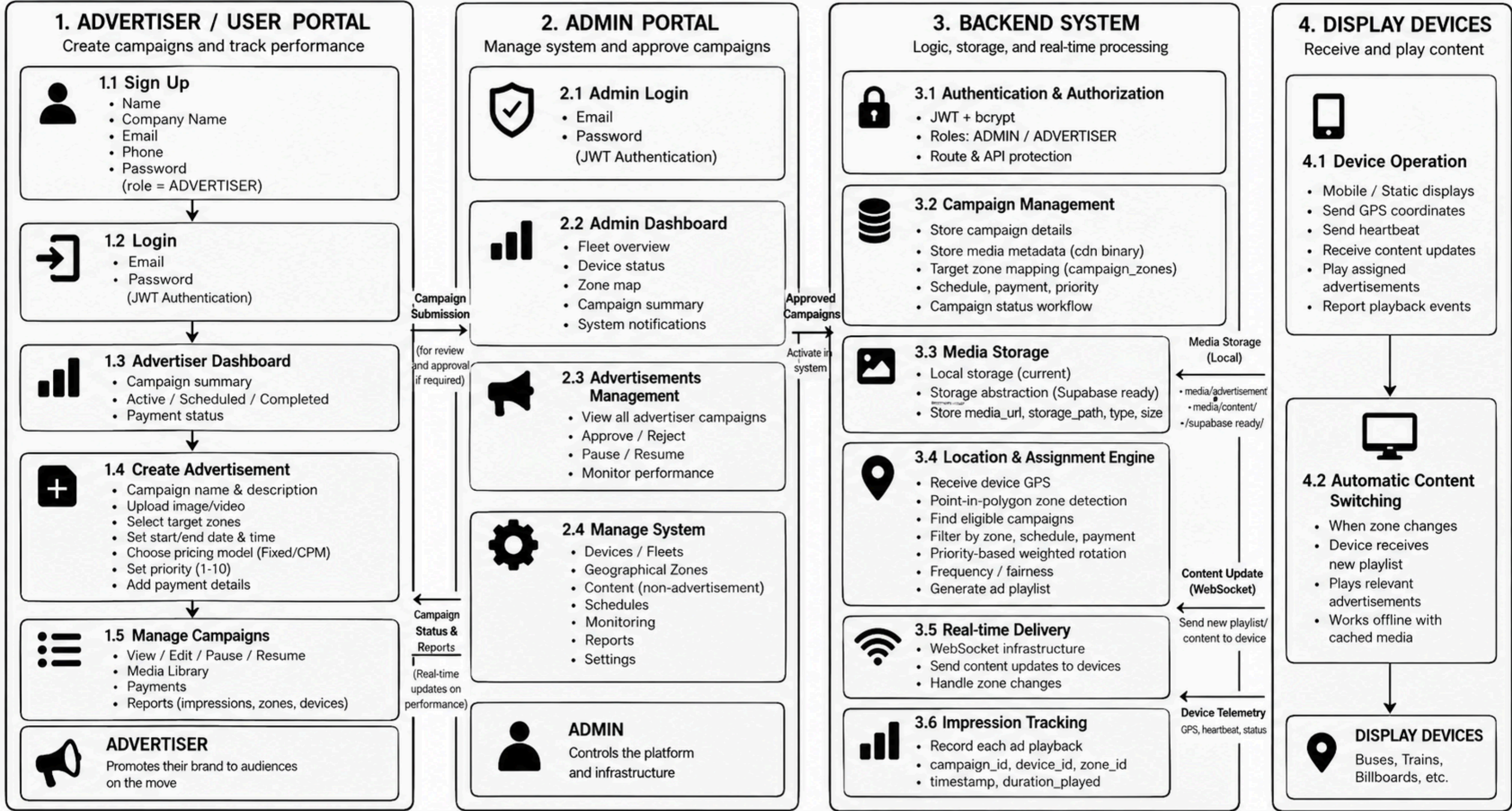
OUR SOLUTION

ADMIN PORTAL

review · confirm
schedule · zones
and devices



SYSTEM ARCHITECTURE*



TECH STACK ↗

Backend / API

High performance API with real-time communication and secure authentication.



Python
(FastAPI)



FastAPI



JWT
(Authentication)



WebSockets
(Realtime API)

Real-Time & Cache

Low-latency communication and fast



Redis
(Cache / Pub-Sub)



WebSocket
(WSS)

Frontend / Admin Panel

Web app for content, device, zone and analytics management.



React



TypeScript



Tailwind CSS



Leaflet
(Maps)

Database

Store application data and spatial/location data.



PostgreSQL
(Primary DB)



PostGIS
(Geospatial Data)



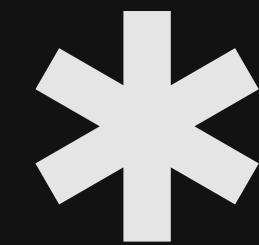
WHAT'S DIFFERENT?



	Usually	LOCATIONSYNC
Roles	Operator uploads for everyone	Advertisers submit; admin confirms before anything goes live
Location	Content tied to a device or group	Server maps GPS → zone → playlist
Where + when	Handled separately	Zone and day/time window checked together (one ad per slot today)
Proof	Schedule shows what was booked	Each play logged: campaign, device, zone, time, duration



FEASIBILITY AND REAL-WORLD USE



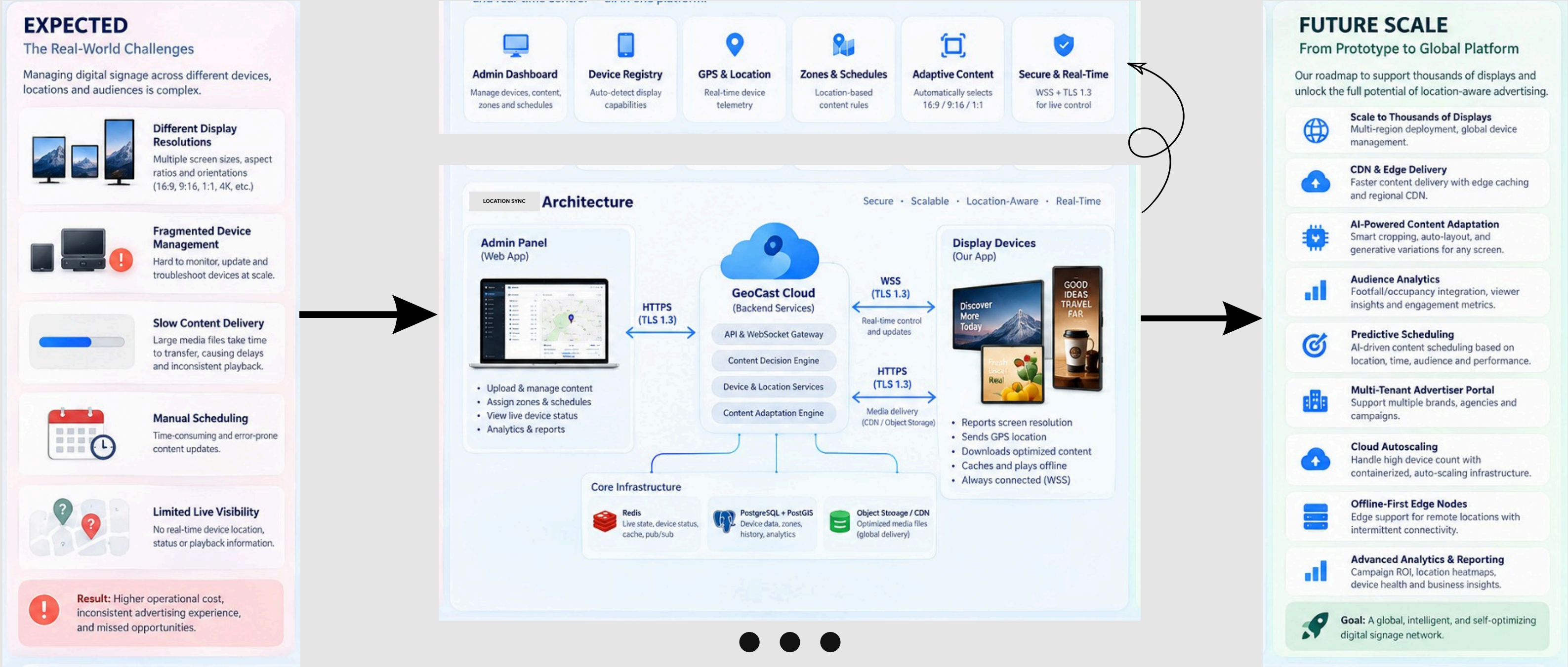
Where it fits

- Bus, train, cab-top screens (mobile)
- Billboard and bus-shelter networks (static)
- Multi-city retail promotions

- * Standard open-source stack
- * One backend, one database, one storage layer runs the full loop
- * Media cached on the device, downloaded once
- * JWT and roles; admin approval before anything goes live



STATUS, LIMITS, ↗ SCALE PATH



THANK *
YOU

TEAM TRACE

